

STRAIGHT LINES

LEVEL - V

SINGLE ANSWER QUESTIONS

- If $\frac{a}{\sqrt{bc}} - 2 = \sqrt{\frac{b}{c}} + \sqrt{\frac{c}{b}}$ where $a, b, c > 0$ then family of lines $\sqrt{a}x + \sqrt{b}y + \sqrt{c} = 0$ passes through the point

A) (1,1) B) (1,-2) C) (-1,2) D) (-1,1)
- The number of integer values of λ for which the x-coordinates of the point of intersection of the lines $3x + 4y = 9$ and $y = \lambda x + 1$ is also an integer is m . Then the area between the lines $y = mx + r, r = 1, 2$ and bounded by coordinate axes is (in sq.units)

A) $\frac{3}{4}$ B) $\frac{4}{9}$ C) $\frac{1}{3}$ D) $\frac{4}{3}$
- Let $2a + 3b + c = 0$. If $x + 5y + 7 = 0$ and $L = 0$ are angle bisectors of two lines $L_1 = 0, L_2 = 0$ and $L = 0$ is a member in the family of lines $ax + by + c = 0$ then area of triangle formed by $L = 0$ with coordinate axes is (in sq.units)

A) 49 B) $\frac{49}{2}$ C) $\frac{49}{5}$ D) $\frac{49}{10}$
- The line $2x + y = 4$ meet x-axis at A and y-axis at B . The perpendicular bisector of AB meets the horizontal line through $(0, -1)$ at C . Let G be the centroid of $\triangle ABC$. The perpendicular distance from G to AB equals

A) $\sqrt{5}$ B) $\frac{\sqrt{5}}{3}$ C) $2\sqrt{5}$ D) $3\sqrt{5}$
- The distance of any point (x, y) from the origin is defined as $d = \max\{|x|, |y|\}$, then the distance of the common point for the family of lines $x(1 + \lambda) + \lambda y + 2 + \lambda = 0$ (λ being parameter) from origin is

A) 1 B) 2 C) $\sqrt{5}$ D) 0
- The distance between two parallel lines is 1 unit. A point 'A' is chosen to lie between the lines at distance 'd' from one of them. Triangle ABC is equilateral with B on one line and C on the other parallel line the length of the side of the equilateral triangle is (in units)

A) $(2/3)\sqrt{d^2 + d + 1}$ B) $2\sqrt{(d^2 - d + 1)}/3$

C) $2\sqrt{(d^2 - d + 1)}$ D) $\sqrt{(d^2 - d + 1)}$
- If the point $P(\alpha^2, \alpha)$ lies in the region corresponding to the acute angle between the lines $x - 3y = 0$ and $x - 5y = 0$ then

A) $\alpha \in (5, 15)$ B) $\alpha \in (5, 8)$

C) $\alpha \in (4, 8)$ D) $\alpha \in (3, 5)$
- The point $(a^2, a+1)$ lies in the angle between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin if

A) $a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$ B) $a \in \left(-3, \frac{1}{3}\right)$

C) $a \in (-\infty, 3) \cup \left(\frac{1}{3}, 1\right)$ D) $a \in \left(\frac{1}{3}, \infty\right)$
- On the portion of the straight line $x + y = 4$ which is intercepted between the axes, a square is constructed away from the origin, with the portion as one of its side. If 'd' denote the perpendicular distance of a side of this square from the origin, then maximum value of 'd' is

A) $2\sqrt{2}$ B) $3\sqrt{2}$ C) $4\sqrt{2}$ D) $6\sqrt{2}$
- The equation of line segment AB is $y = x$. If A & B lie on same side of line mirror $2x - y = 1$, then the equation of image of AB

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with respect to line mirror $2x - y = 1$ is

- A) $y = 7x - 5$ B) $y = 7x - 6$
C) $y = 3x - 7$ D) $y = 6x - 5$

11. A line passing through P (6, 4) meets the coordinates axes at A and B respectively. If O is the origin, then locus of the centre of the circumcircle of triangle OAB is

- A) $3x^{-1} + y^{-1} = 1$ B) $x^{-1} + 2y^{-1} = 1$
C) $x^{-1} + y^{-1} = 1$ D) $3x^{-1} + 2y^{-1} = 1$

12. Let $ax + by + c = 0$ be a variable straight line, where a, b and c are 1st, 3rd and 7th terms of some increasing A.P. Then the variable straight line always passes through a fixed point which lies on

- A) $x^2 + y^2 = 13$ B) $x^2 + y^2 = 5$
C) $y^2 = 9x$ D) $3x + 4y = 9$

13. Line L has intercepts a and b on the coordinate axes. When the axes are rotated through a fixed given angle keeping the origin fixed, the same line L has intercepts p and q, then [IIT 1990]

- A) $a^2 + b^2 = p^2 + q^2$
B) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$
C) $a^2 + p^2 = b^2 + q^2$
D) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{b^2} + \frac{1}{q^2}$

14. Given the family of lines,

$a(2x + y + 4) + b(x - 2y - 3) = 0$. Among the lines of the family, the number of lines situated at a distance of $\sqrt{10}$ from the point M(2, -3) is

- A) 0 B) 1 C) 2 D) ∞

15. If the angle between the lines represented by $6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0$ is $\tan^{-1}(m)$ and $a^2 + b^2 - ab - a - b + 1 \leq 0$ then $5a + 6b$ is equal to

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- A) m B) $\frac{1}{m}$ C) 2m D) $\frac{1}{2m}$

16. PQR is an equilateral triangle such that the vertices Q and R lie on the lines $x + y = \sqrt{2}$ and $x + y = 7\sqrt{2}$ respectively. If P lies between the two lines at a distance 4 from one of them then the length of side of equilateral triangle PQR is (in units)

- A) 8 B) $\frac{4\sqrt{7}}{\sqrt{3}}$ C) $\frac{\sqrt{85}}{3}$ D) $\frac{4\sqrt{5}}{\sqrt{3}}$

17. P(m, n) ($m, n \in \mathbb{N}$) is any point in the interior of the quadrilateral formed by the pair of lines $xy = 0$ and two lines $2x + y - 2 = 0$ and $4x + 5y = 20$ then the possible number of positions of the points P is

- A) 4 B) 5 C) 6 D) 11

18. If d_1, d_2 denotes the lengths of the perpendiculars from the point (2, 3) on the lines given by $15x^2 + 31xy + 14y^2 = 0$. If

$d_1 > d_2$ then $d_1^2 - d_2^2 + \frac{1}{74} + \frac{1}{13} =$

- A) -2 B) 0 C) 2 D) 3

19. Two of the straight lines given by $3x^3 + 3x^2y - 3xy^2 + py^3 = 0$ are at right angles. The equation of line passing through (1, 1) and perpendicular to $y = px$ is

- A) $x + 3y - 4 = 0$ B) $3x - y - 2 = 0$
C) $3x + y - 4 = 0$ D) $x - 3y + 2 = 0$

20. ABCD is a square whose vertices A (0,0) B (2,0), C (2,2), D (0, 2). This square is rotated in the xy-plane with an angle of 30° in anticlockwise direction about an axis passing through the vertex A, the equation of the diagonal BD of this rotated square is

- A) $\sqrt{3}x + (1 - \sqrt{3})y = \sqrt{3}$
B) $(1 + \sqrt{3})x - (1 - \sqrt{2})y = 2$

$$C) (2 - \sqrt{3})x + y = 2(\sqrt{3} - 1)$$

$$D) \sqrt{3}x + (1 - \sqrt{2})y = \sqrt{3} - 1$$

21. The equations to a pair of opposite sides of parallelogram are $x^2 - 5x + 6 = 0$ and $y^2 - 6y + 5 = 0$, the equations to its diagonals are [IIT 1994]

$$A) x + 4y = 13, y = 4x - 7$$

$$B) 4x + y = 13, 4y = x - 7$$

$$C) 4x + y = 13, y = 4x - 7$$

$$D) y - 4x = 13, y + 4x = 7$$

22. Two sides of a triangle having the joint equation $(x - 3y + 1)(x + y - 2) = 0$ the third side which is variable always passes through the point $(-5, -1)$, then possible values of slope of the third side such that origin is an interior point of triangle are..

$$A) \left(-1, \frac{-1}{5}\right) \quad B) \left(\frac{-1}{5}, -1\right)$$

$$C) (0, 2) \quad D) \mathbb{R}$$

23. Through the point $P(3, 4)$ a pair of perpendicular lines are drawn which meet x -axis at the points A and B . The locus of incentre of triangle PAB is

$$A) x^2 - y^2 - 6x - 8y + 25 = 0$$

$$B) x^2 + y^2 - 6x - 8y + 25 = 0$$

$$C) x^2 - y^2 + 6x + 8y + 25 = 0$$

$$D) x^2 + y^2 + 6x + 8y + 25 = 0$$

24. Two parallel lines lying in the same quadrant make intercepts a and b on x, y axes respectively between them, then the distance between the lines is

$$A) \frac{ab}{\sqrt{a^2 + b^2}} \quad B) \sqrt{a^2 + b^2}$$

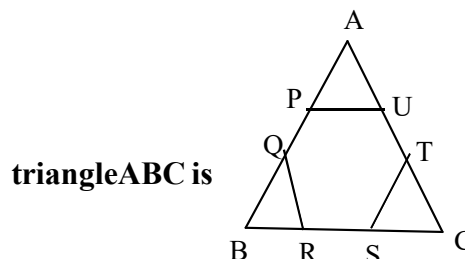
$$C) \frac{1}{\sqrt{a^2 + b^2}} \quad D) \frac{1}{a^2} + \frac{1}{b^2}$$

25. The four sides of a quadrilateral are given by $xy(x-2)(y-3) = 0$. The equation of the line parallel to $x - 4y = 0$ that divides the quadrilateral into two equal areas is

$$A) x - 4y + 7 = 0 \quad B) x - 4y + 5 = 0$$

$$C) x - 4y + 11 = 0 \quad D) x - 4y + 3 = 0$$

26. Each side of triangle ABC is divided into three equal parts as shown in the figure ratio of area of hexagon $PQRSTU$ to area of



triangle ABC is

$$A) 5/9 \quad B) 2/3 \quad C) 1/2 \quad D) 3/4$$

27. A line through $P(3, 4)$ cuts the lines $x = 6$ and $y = 8$ at L and M respectively. Q is a variable point on the line such that

$$\frac{1}{PQ} = \frac{1}{PL} + \frac{1}{PM} \text{ then the locus of } Q \text{ is}$$

$$A) 4x + 3y - 36 = 0 \quad B) x^2 + y^2 = 36$$

$$C) 3x - 4y - 36 = 0 \quad D) 4x^2 - 9y^2 = 36$$

28. $ABCDEF$ is a regular hexagon in anticlockwise sense and $A(2, 0)$, $B(4, 0)$ then the coordinates of C are

$$A) \left(\frac{9}{2}, \sqrt{3}\right) \quad B) (5, \sqrt{3})$$

$$C) \left(\frac{5}{2}, \frac{\sqrt{3}}{2}\right) \quad D) \left(2, \frac{\sqrt{3}}{2}\right)$$

29. Lines $(1 + \lambda)x + (4 - \lambda)y + (2 + \lambda) = 0$

and $(4 - \lambda)x - (1 + \lambda)y + (6 - 3\lambda) = 0$ are concurrent at points A and B respectively and intersect at C then locus of centroid of ΔABC is (λ is parameter)

$$A) \left(x + \frac{3}{2}\right)^2 + \left(y + \frac{7}{10}\right)^2 = \frac{17}{50}$$

$$B) \left(x - \frac{3}{2}\right)^2 + \left(y - \frac{7}{10}\right)^2 = \frac{17}{50}$$

$$C) \left(x - \frac{3}{2}\right)^2 + \left(y + \frac{7}{10}\right)^2 = \frac{17}{450}$$

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$$D) \left(x + \frac{3}{2}\right)^2 + \left(y + \frac{7}{10}\right)^2 = \frac{17}{450}$$

30. The line $x + y = 1$ meets x -axis at A and y -axis at B. P is the midpoint of AB. P_1 is the foot of the perpendicular from P to OA; M_1 is that from P_1 to OP; P_2 is that from M_1 to OA and so on. If P_n denotes the n th foot of the perpendicular on OA from M_{n-1} then OP_n

A) $1/2$ B) $1/2^n$ C) $1/2^{n/2}$ D) $1/\sqrt{2}$

31. A ray of light leaves the point (3, 4) reflects off the y -axis towards x -axis and again after reflecting from x -axis finally arrives at the point (8, 2) then the abscissa of point where the reflected ray meets x -axis is

A) $\frac{9}{2}$ B) $\frac{13}{3}$ C) $\frac{14}{3}$ D) $\frac{16}{3}$

32. If the lengths of the medians through acute angles of a right angled triangle are 3 and 4 then the area of the triangle is (in sq. units)

A) $\frac{4}{3}\sqrt{11}$ B) $\frac{2}{3}\sqrt{11}$ C) $2\sqrt{11}$ D) $3\sqrt{11}$

33. A line is drawn through the point (-4, 5) such that the distance of the point (-3, 2) from the line is d then the maximum value of d

A) 0 B) $\sqrt{10}$ C) $\sqrt{\frac{5}{2}}$ D) $\sqrt{5}$

34. A lattice point in a plane is a point for which both coordinates are integers. The number of lattice points inside the triangle whose sides are $x = 0$, $y = 0$ and $9x + 223y = 2007$ is

A) 198 B) 173 C) 99 D) 888

35. A point P (x, y) moves such that the sum of its distances from the lines $2x - y - 3 = 0$ and $x + 3y + 4 = 0$ is 7. The area bounded by locus of P is (in sq. units)

A) 70 B) $70\sqrt{2}$ C) $35\sqrt{2}$ D) 140

36. A triangle ABC right angled at C moves such that A and B always lie on the positive x and y -axes then locus of C is

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- A) Straight Line B) Circle
C) Parabola D) Ellipse

37. The slopes of sides of a triangle are -1, -2, 3. If the orthocentre of the triangle is the origin

O; then the locus of its centroid is $\frac{y}{x} =$

A) $\frac{2}{3}$ B) $\frac{2}{5}$ C) $\frac{2}{7}$ D) $\frac{2}{9}$

38. Let α, β be the roots of $ax^2 + 2hx + b = 0$ and γ, δ be the roots of $a_1x^2 + 2h_1x + b_1 = 0$. Consider the points

$A(\alpha, 0)$ $B(\beta, 0)$ $C(\gamma, 0)$ $D(\delta, 0)$.

If the sum of ratios in which C and D

Divides AB is zero then $ab_1 + a_1b =$

A) 0 B) hh_1 C) $2hh_1$ D) -1

39. Let PQR be right angled isosceles triangle right angled at P(2, 1). If the equation of the line QR is $2x + y = 3$, then the equation representing the pair of lines PQ and PR is [IIT 1999]

A) $3x^2 - 3y^2 + 8xy + 20x + 10y + 25 = 0$

B) $3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$

C) $3x^2 - 3y^2 + 8xy + 20x + 15y + 25 = 0$

D) $3x^2 - 3y^2 + 8xy - 10x - 15y - 25 = 0$

40. Let PS be the median of the triangle with vertices P(2, 2), Q(6, -1) and R(7, 3). The equation of the line passing through (1, -1) and parallel to PS is [IIT 2000]

A) $2x - 9y - 7 = 0$ B) $2x - 9y - 11 = 0$

C) $2x + 9y - 11 = 0$ D) $2x + 9y + 7 = 0$

41. The number of integer values of m , for which the x -coordinate of the point of intersection of the lines $3x + 4y = 9$ and $y = mx + 1$ is also an integer [IIT 2001]

A) 2 B) 0 C) 4 D) 1

42. Area of the parallelogram formed by the lines $y = mx$, $y = mx + 1$, $y = nx$ and $y = nx + 1$ equals. (in sq units) [IIT 2001]

A) $\frac{|m+n|}{(m-n)^2}$ B) $\frac{2}{|m+n|}$

C) $\frac{1}{|m+n|}$ D) $\frac{1}{|m-n|}$

43. Let $P = (-1, 0)$, $Q = (0, 0)$ and $R = (3, 3\sqrt{3})$ be three points. Then the equation of the bisector of the angle PQR in [IIT 2002]

A) $\frac{\sqrt{3}}{2}x + y = 0$ B) $x + \sqrt{3}y = 0$

C) $\sqrt{3}x + y = 0$ D) $x + \frac{\sqrt{3}}{2}y = 0$

44. A straight line through the origin O meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively. Then the point O divides the segment PQ in the ratio [IIT 2002]

A) 1 : 2 B) 3 : 4 C) 2 : 1 D) 4 : 3

45. The number of integral points (integral point means both the coordinates should be integers) exactly in the interior of the triangle with vertices $(0, 0)$, $(0, 21)$ and $(21, 0)$ is [IIT 2003]

A) 133 B) 190 C) 233 D) 105

46. m, n are integer with $0 < n < m$. A is the point (m, n) on the cartesian plane. B is the reflection of A in the line $y = x$. C is the reflection of B in the y -axis, D is the reflection of C in the x -axis and E is the reflection of D in the y -axis. The area of the pentagon $ABCDE$ is

A) $2m(m+n)$ B) $m(m+3n)$

C) $m(2m+3n)$ D) $2m(m+3n)$

47. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ . The point R inside the triangle OPQ is such that the triangles OPR , PQR , OQR are of equal area. The coordinates of R are [IIT 2007]

A) $\left(\frac{4}{3}, 3\right)$ B) $\left(3, \frac{2}{3}\right)$ C) $\left(3, \frac{4}{3}\right)$ D) $\left(\frac{4}{3}, \frac{2}{3}\right)$

48. A straight line L through the point $(3, -2)$ is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$, if L also intersects the x -axis,

then the equation of L is [IIT 2011]

A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

MULTIPLE ANSWER QUESTIONS

49. One of the bisectors of the angle between the lines

$$a(x-1)^2 + 24(x-1)(y-2) + b(y-2)^2 = 0$$

is $x + 2y - 5 = 0$ then the other bisector

A) $2x - y = 0$ B) $2x - y - 1 = 0$

C) Passes through origin

D) Passes through $(1, 1)$

50. For all values of θ , the lines represented by the equation

$$(2 \cos \theta + 3 \sin \theta)x + (3 \cos \theta - 5 \sin \theta)y$$

$$-(5 \cos \theta - 2 \sin \theta) = 0$$

A) Pass through a fixed point

B) Pass through the point $(1, 1)$

C) Pass through a fixed point whose reflection in

the line $x + y = \sqrt{2}$ is $(\sqrt{2} - 1, \sqrt{2} - 1)$

D) Pass through the origin

51. A line through $A(-5, -4)$ with slope $\tan \theta$ meets the lines $x + 3y + 2 = 0$, $2x + y + 4 = 0$, $x - y - 5 = 0$ at B , C , D respectively, such that

$$\left(\frac{15}{AB}\right)^2 + \left(\frac{10}{AC}\right)^2 = \left(\frac{6}{AD}\right)^2 \text{ then}$$

A) $\frac{15}{AB} = \cos \theta + 3 \sin \theta$

B) $\frac{10}{AC} = 2 \cos \theta + \sin \theta$

C) $\frac{6}{AD} = \cos \theta - \sin \theta$

D) Slope of the line is $-\frac{2}{3}$

52. The lines L_1 and L_2 denoted by

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$$3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$$

intersects at the point P and having slopes m_1 and m_2 respectively. The acute angle between them is θ . Which of the following relations hold good.

A) $m_1 + m_2 = \frac{5}{4}$ B) $m_1 m_2 = \frac{3}{8}$

C) $\theta = \sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$

D) Sum of abscissa and ordinate of point P is -1

53. Two sides of a rhombus ABCD are parallel to lines $y = x + 2$ and $y = 7x + 3$. If the diagonals of the rhombus intersect at point (1, 2) and the vertex A is on the y-axis is, then the possible coordinates of A are

A) $\left(0, \frac{5}{2}\right)$ B) (0, 0) C) (0, 5) D) (0, 3)

54. Two lines from the family of lines $(1 + 2\lambda)x - (1 + \lambda)y + 1 = 0$ and the line $x + y = 5$ form an equilateral triangle. the equation of the two lines can be

A) $y - 2 = (2 + \sqrt{3})(x - 1)$

B) $y - 2 = (2 - \sqrt{3})(x - 1)$

C) $y - 2 = (3 - \sqrt{2})(x - 1)$

D) $y - 1 = (3 - \sqrt{2})(x - 1)$

55. Let x_1 and y_1 be the roots of $x^2 + 8x - 2009 = 0$; x_2 and y_2 be the roots of $3x^2 + 24x - 2010 = 0$ and x_3 and y_3 be the roots of $9x^2 + 72x - 2011 = 0$. The points

$A(x_1, y_1)$ $B(x_2, y_2)$ and $C(x_3, y_3)$

- A) can not lie on a circle
B) form a triangle of area 2 sq. units
C) form a right angled triangle
D) are collinear

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56. The sides of a triangle are the straight lines $x + y = 1$, $7y = x$ and $\sqrt{3}y + x = 0$. Then which of the following is an interior point of the triangle

- A) Circumcentre B) Centroid
C) Incentre D) orthocentre

57. A ray travelling along the line $3x - 4y = 5$ after being reflected from a line l travel along the line $5x + 12y = 13$. Then the equation of the line l is

- A) $x + 8y = 0$ B) $x = 8y$
C) $32x + 4y = 65$ D) $32x - 4y + 65 = 0$

58. All the point lying inside the triangle formed by the points (1, 3), (5, 6) and (-1, 2) satisfy

- A) $3x + 2y \geq 0$ B) $2x + y + 1 \geq 0$
C) $-2x + 11 \geq 0$ D) $2x + 3y - 12 \geq 0$

59. Two straight lines $u = 0$ and $v = 0$ passes the origin and the angle between them is $\tan^{-1}\left(\frac{7}{9}\right)$.

If the ratio of slopes of $v = 0$ and $u = 0$ is $\frac{9}{2}$,

then their equations are

- A) $y = 3x$ and $3y = 2x$
B) $2y = 3x$ and $3y = x$
C) $y + 3x = 0$ and $3y + 2x = 0$
D) $2y + 3x = 0$ and $3y + x = 0$

60. A ray of light travelling along the line $x + y = 1$ is incident on the x-axis and after refraction it enters the other side of the x-axis by turning $\pi/6$ away from the x-axis. The equation of the line along which the refracted ray travels is

- A) $x + (2 - \sqrt{3})y = 1$
B) $(2 - \sqrt{3})x + y = 1$
C) $y + (2 + \sqrt{3})x = 2 + \sqrt{3}$

D) $y + (2 - \sqrt{3})x = 2 - \sqrt{3}$

COMPREHENSION QUESTIONS

PASSAGE : I

Consider 3 non-collinear points $A(9,3)$, $B(7,-1)$, and $C(1,-1)$. Let $P(a, b)$ be the centre and R is the radius of circle 'S' passing through points A, B, C . Also $H(\bar{x}, \bar{y})$ are the coordinates of the orthocentre of triangle ABC whose area be denoted by Δ .

61. If D, E and F are the middle points BC, CA and AB respectively then the area of the triangle DEF is :
A) 12 B) 6 C) 4 D) 3
62. The value of $a + b + R$ equals
A) 3 B) 12
C) 13 D) None of these
63. The ordered pair $(\bar{x}, \bar{y})$ is :
A) (9, 6) B) (9, -6) C) (9, -5) D) (9, 5)

PASSAGE : II

The equation of an altitude of an equilateral triangle is $\sqrt{3}x + y = 2\sqrt{3}$ and one of its vertices is $(3, \sqrt{3})$ then

64. The possible number of triangle(s) is
A) 1 B) 2 C) 3 D) 4
65. Which of the following can not be the vertex of the triangle
A) $(0, 0)$ B) $(0, 2\sqrt{3})$
C) $(3, -\sqrt{3})$ D) $(2\sqrt{3}, 0)$
66. Which of the following can be the possible orthocentre of the triangle
A) $(1, \sqrt{3})$ B) $(0, \sqrt{3})$ C) $(0, 2)$ D) $(0, 3)$

PASSAGE : III

Let $A(0,0)$, $B(5,0)$, $C(5,3)$ and $D(0,3)$ be the vertices of rectangle $ABCD$, If P is a variable point lying inside the rectangle $ABCD$ and $d(P, L)$ denotes perpendicular

distance of point P from line L

67. If $d(P, AB) \leq \min\{d(P, BC), d(P, CD), d(P, AD)\}$, then area of the region in which P lies is (in sq.units)
A) $\frac{17}{4}$ B) $\frac{19}{4}$ C) $\frac{21}{4}$ D) $\frac{23}{4}$
68. If $d(P, AB) \geq \max[d(P, BC), d(P, CD), d(P, AD)]$, then the area of the region in which P lies is (in sq.units)
A) 1 B) $1/2$ C) $3/4$ D) $1/4$
69. If $\left(d(P, AB) - \frac{3}{2}\right)^2 + (d(P, AD))^2 \geq 1$, then area of region in which P lies is (in sq.units)
A) $15 - 2\pi$ B) $10 - \frac{\pi}{2}$
C) $15 - \pi$ D) $15 - \frac{\pi}{2}$

PASSAGE : IV

The line $6x + 8y = 48$ intersects the coordinate axes at A and B respectively. A line L bisects the area and perimeter of triangle OAB where 'O' is origin

70. The number of possible such lines is
A) 1 B) 2
C) 3 D) more than 3
71. The slope of the line L can be
A) $\frac{10 + 5\sqrt{6}}{10}$ B) $\frac{10 - 5\sqrt{6}}{10}$
C) $\frac{8 + 3\sqrt{6}}{10}$ D) $\frac{8 - 3\sqrt{6}}{10}$
72. The line L does not intersect the side ____ of the triangle OAB
A) AB B) OB
C) OA D) can intersect all sides

MATRIX-MATCHING QUESTIONS

73. A line cuts x -axis at A and y -axis at B such that $AB = \ell$. Match the following Column I

STRAIGHT LINES

- A) Circumcentre of ΔOAB
- B) Orthocentre of ΔOAB
- C) Incentre of the ΔOAB
- D) Centroid of the ΔOAB

Column II (Locus of the point lies on)

p) $x^2 + y^2 = \frac{\ell^2}{9}$

q) $x^2 + y^2 = \frac{\ell^2}{4}$

r) $x^2 + y^2 = 0$

s) $y = x$

74. **Vertex A of the triangle ABC is at origin. The equation of medians through B and C are $15x - 4y - 240 = 0$ and**

$15x - 52y + 240 = 0$ respectively

Column I

- A) The coordinate of incentre of triangle ABC are
- B) The coordinates of centroid of triangle ABC are
- C) The coordinates of excentre opposite to vertex C of triangle ABC are
- D) The coordinates of orthocentre of triangle ABC are

Column II

p) $\left(\frac{56}{3}, 10\right)$

q) $(21, 12)$

r) $(12, 21)$

s) $(-4, 7)$

t) $(0, 63)$

75. **Column I**

- A) The number of integral values 'a' for which point (a, a^2) lies completely inside the triangle $x = 0, y = 0, 2y + x = 3$.

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- B) The number of values of a of the form $\frac{k}{3}$

where $k \in I$ so that point (a, a^2) lies between the lines $x + y = 2$ and $4x + 4y - 3 = 0$

- C) The reflection of point $(t - 1, 2t + 2)$ in a line is $(2t + 1, t)$ then the slope of the line is

- D) In a triangle ABC, the bisector of angles B and C lies along the lines $y = x$ and $y = 0$. If A is $(1, 2)$ then $\sqrt{10} d(A, BC)$ equals (where $d(A, BC)$ denotes the perpendicular distance of A from BC)

Column II

p) 0

q) 1

r) 2

s) 4

76. **Column I**

- A) If a, b, c are in A.P, then lines $ax + by + c = 0$ are concurrent
- B) A point on the line $x + y = 4$ which lies at a unit distance from the line $4x + 3y = 10$ is
- C) Orthocentre of triangle made by lines $x + y = 1, x - y + 3 = 0, 2x + y = 7$ is
- D) Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If orthocentre is the origin then coordinate of the third vertex are

Column II

p) $(-4, -7)$

q) $(-7, 11)$

r) $(1, -2)$

s) $(-1, 2)$

t) $(3, 1)$

77. **Column I**

- A) If the lines $x + 2ay + a = 0, x + 3by + b = 0, x + 4cy + c = 0$, where a, b, c $\in R$ are concurrent, then a, b, c are in

B) The point with coordinates $(2a, 3a), (3b, 2b), (c, c)$ where $a, b, c \in \mathbb{R}$ are collinear, then a, b, c are in

C) If lines $ax + 2y + 1 = 0, bx + 3y + 1 = 0$ and $cx + 4y + 1 = 0$ where $a, b, c \in \mathbb{R}$ passes through the same point, then a, b, c are in

D) Let a, b, c be distinct non-negative real numbers. If the lines $ax + ay + c = 0, x + 1 = 0, cx + cy + b = 0$ pass through the same point then a, b, c are in

Column II

p) A.P.

q) G.P

r) H.P

s) Neither A.P. nor G.P. nor H.P

INTEGER QUESTIONS

78. If the co-ordinates of the orthocentre of the triangle formed by the lines $y = 0, 37x - 36y + 37 \times 36 = 0$ and $64x - 63y + 64 \times 63 = 0$ is (a, b) then $a - b - 4535$ is
79. A straight line passing through the point $(87, 33)$ cuts the positive direction of the co-ordinate axes at the point P and Q. If O is the origin then minimum area of the triangle OPQ is P then P - 5740 is
80. If $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$ represents a pair of straight lines, then λ^2 is
81. $A(-3, 2), B(2, 5)$ be two points. If the point P(O, K) on y - axis, such that $|PA - PB|$ is greatest, then $\frac{K-1}{5}$ is equal to
82. A straight line L through the origin meets the line $x + y = 1$ and $x + y = 3$ at P and Q respectively. Through P and Q two straight lines L_1 and L_2 are drawn, parallel to $2x - y = 5$ and $3x + y = 5$ respectively. Lines L_1 and L_2 intersect at R. If the locus of R, as L varies, is a straight line $x - 3y + \lambda = 0$, then $\lambda =$
83. P is a point in the plane of triangle ABC

such that $\text{area}(\triangle PAB) = \text{area}(\triangle PBC)$

$= \text{area}(\triangle PCA)$ then the number of possible positions of P is

84. OPQR is square ('O' being origin) and M, N are middle points of sides PQ, QR respectively and the ratio of areas of

square and triangle OMN is $\frac{P}{q}$ (where

P, q are relatively prime) then $P - q$ is

85. Number of lines which are equidistant from the three vertices of a triangle is
86. A and B are two fixed points such that $AB = 3$ units. P is a point such that

$\frac{PA}{PB} = 2$ then the maximum area of

triangle PAB is (in sq.units)

87. Each side of a square is of length 4 units. The centre of the square is at $(3, 7)$ and one of the diagonals is parallel to the line $y = x$. If the vertices of the square be

$(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4) , then

$\max(y_1, y_2, y_3, y_4) - \min(x_1, x_2, x_3, x_4)$ is

88. Let the algebraic sum of the perpendicular distances from the points $(2, 0), (0, 2)$ and $(1, 1)$ to a variable straight line be zero; then the line passes through a fixed point whose coordinates are (a, b) , then $a + b$ is [IIT 1991]

89. Two rays in the first quadrant $x + y = |a|$ and $ax - y = 1$ intersect each other in the interval $a \in (a_0, \infty)$, the value of a_0 is [IIT 2006]

90. B $(2, -7)$ is vertex of a triangle. Equation of altitude and median drawn from different vertices are $3x + y + 11 = 0$ and $x + 2y + 7 = 0$ respectively. If the equation of

side AC is $7x + 9y + \lambda = 0$ then $\frac{\lambda + 1}{4} =$

91. P is a point on the circumcircle of an equilateral triangle ABC of side length $\sqrt{3}$ units then the value of

$PA^2 + PB^2 + PC^2$ is

STRAIGHT LINES

92. The length of the largest altitude of the triangle formed by the lines $7x+2y+10=0$, $7x+2y-10=0$ and $y+2=0$ is
93. For any given line $15x+8y=34$ minimum value of $x^2 + y^2$ is
94. If $A(n, n^2)$ (where $n \in N$) is any point in the interior of the quadrilateral formed by $x=0$, $y=0$, $3x+y-4=0$ and $4x+y-21=0$ then the possible number of positions of A is
95. In a square ABCD, $G_1=(3,5)$ and $G_2=(5,3)$ are the centroids of $\triangle ABC$ and $\triangle ACD$ then side of the square is (in units)
96. The number of points on the line $3x+4y=5$ which are at a distance of $\sec^2 \theta + 2 \cos \sec^2 \theta$ (for admissible values of θ) from the point $(1, 3)$ is

ASSERTION & REASON

Each question contains STATEMENT – 1 (Assertion) and STATEMENT – 2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which ONLY ONE is correct.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, statement-2 is True

97. Statement-1: The coordinates of the point which divides the line joining $(5, -2)$ and $(9, 6)$ externally in the ratio $3 : 1$ are $(8, 4)$. Statement-2: If the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

represents a pair of parallel lines then

$$\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$$

98. Statement-1: The point $(-4, 1)$ lies on the non-origin side of the line $x - 4y + 2 = 0$

Statement-2: When point P (x_1, y_1) lie above the line $ax+by+c=0$ ($b \neq 0$) then

$$\frac{ax_1 + by_1 + c}{b} > 0$$

99. Statement-1: Four non-collinear points on the x-y plane are A(2, 3), B(3, 5), C(7, 7) and D(4, 5).

Statement-2: The point D is the centroid of the triangle ABC.

100. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q, respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

Statement-1 : The ratio PR : RQ equals

$$2\sqrt{2} : \sqrt{5}.$$

Because

Statement-2 : In any triangle, bisector of an angle divides the triangle into two similar triangles. [IIT 2007]

KEY
LEVEL -V

SINGLE ANSWER QUESTIONS

- | | | | |
|-------|-------|-------|-------|
| 1) D | 2) A | 3) D | 4) A |
| 5) B | 6) B | 7) D | 8) A |
| 9) D | 10) B | 11) D | 12) A |
| 13) B | 14) B | 15) C | 16) B |
| 17) C | 18) C | 19) D | 20) C |
| 21) C | 22) A | 23) A | 24) A |
| 25) B | 26) B | 27) A | 28) B |
| 29) D | 30) B | 31) B | 32) A |
| 33) B | 34) D | 35) B | 36) A |
| 37) D | 38) C | 39) B | 40) D |
| 41) A | 42) D | 43) C | 44) B |
| 45) B | 46) B | 47) C | 48) B |

MULTIANSWER QUESTIONS

- | | |
|-------------|-----------|
| 49) A,C | 50) A,B,C |
| 51) A,B,C,D | 52) B,C,D |
| 53) A,B | 54) A,B |
| 55) A,D | 56) B,C |
| 57) B,C | 58) A,B,C |
| 59) A,B,C,D | 60) A,C |

COMPREHENSION QUESTIONS

- | | | | |
|-------|-------|-------|-------|
| 61) D | 62) B | 63) C | 64) B |
| 65) D | 66) A | 67) C | 68) D |
| 69) D | 70) A | 71) B | 72) C |

MATRIX MATCH QUESTIONS

73) A-q; B-r; C-s; D-p

74) A-q; B-p; C-s; D-t

75) A-p; B-r; C-q; D-s

76) A-r; B-q, t; C-s; D-p

77) A-r; B-s; C-p; D-q

INTEGER QUESTIONS

78) 1 79) 2 80) 4 81) 2

82) 5 83) 4 84) 5 85) 3

86) 3 87) 8 88) 2 89) 1

90) 5 91) 6 92) 7 93) 4

94) 1 95) 6 96) 2

ASSERTION & REASON QUESTIONS

97) D 98) B 99) A 100) C

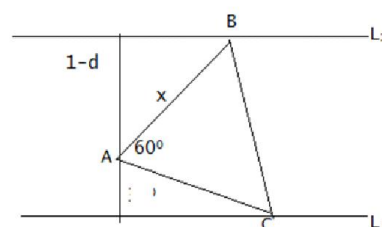
HINTS
LEVEL - V

SINGLE ANSWER QUESTIONS

- $a - 2\sqrt{bc} = b + c$
 $\Rightarrow (\sqrt{b} + \sqrt{c})^2 - (\sqrt{a})^2 = 0$
 or $\sqrt{b} + \sqrt{c} - \sqrt{a} = 0$ ($\because \sqrt{b} + \sqrt{c} + \sqrt{a} \neq 0$)
 $\therefore \sqrt{a}x + \sqrt{b}y + \sqrt{c}$ passes through the fixed point $(-1, 1)$
- x - coordinate of point of intersection of
 $3x + 4y = 9$ and $y = \lambda x + 1$ is $\frac{5}{3 + 4\lambda}$
 Which is an integer if $\lambda = -1, \lambda = -2$
 $\therefore m = 2$
 Two lines are $y = 2x + 1, y = 2x + 2$
 Area of the portion $= 1 - \frac{1}{4} = \frac{3}{4}$ sq. units
- Point of concurrence of $ax + by + c = 0$ is $(2, 3)$
 Slope of L is $\frac{-1}{\left(-\frac{1}{5}\right)} = 5$
 Equation of line L = 0 is
 $y - 3 = 5(x - 2) \Rightarrow$
 $5x - y - 7 = 0$

$$\text{Area of triangle} = \frac{(7)^2}{2|5(-1)|} = \frac{49}{10} \text{ sq. units}$$

- Slope of CD is $\frac{1}{2} \Rightarrow C \equiv (-5, -1)$
 Perpendicular distance from G to AB =
 $\frac{1}{3}$ (perpendicular distance from C to AB).
- Fixed point, $p \equiv (-2, 1)$
 Distance of P from origin = 2



6.

$$\text{From the figure } x \cos(\theta + 30^\circ) = d$$

$$(i) \quad x \sin \theta = 1 - d$$

(ii) Dividing (i) by (ii), we have

$$\sqrt{3} \cot \theta = \frac{1+d}{1-d}$$

eliminate θ

7. P lies in the acute angle

$$\Rightarrow \alpha^2 - 3\alpha > 0 \text{ and } \alpha^2 - 5\alpha < 0$$

$$\Rightarrow \alpha \in (-\infty, 0) \cup (3, \infty) \text{ and } \alpha \in (0, 5)$$

$$\therefore \alpha \in (3, 5)$$

8. Intersection of $x = (y - 1)^2$ with lines are

$$(y - 1)^2 = \frac{y - 1}{3} \Rightarrow y = 1, y = \frac{4}{3}$$

$$\Rightarrow p = \left(\frac{1}{9}, \frac{4}{3}\right)$$

$$(y - 1)^2 = 5 - 2y \Rightarrow y^2 = 4 \Rightarrow y = \pm 2$$

$$Q \equiv (1, 2), R \equiv (9, -2)$$

$$\Rightarrow a + 1 \in (-2, 1) \cup \left(\frac{4}{3}, 2\right)$$

STRAIGHT LINES

9. Distance from origin to $x + y - 4 = 0$ is

$$\frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$\text{Side length} = \sqrt{16+16} = 4\sqrt{2}$$

$$\text{Max distance} = 2\sqrt{2} + 4\sqrt{2} = 6\sqrt{2}$$

10. Image of $(0,0)$ w.r.t. L lies on L_1 .

$$\text{Image of } (0,0) \text{ w.r.t. } L \text{ is } \left(\frac{4}{5}, \frac{-2}{5}\right)$$

$$L_1 \text{ passes through } (1,1) \text{ and } \left(\frac{4}{5}, \frac{-2}{5}\right).$$

11. Let mid point of $AB = (x_1, y_1) = M$

$$A = (2x_1, 0), B(0, 2y_1)$$

$$\text{Equation of line } AB \text{ is } \frac{x}{2x_1} + \frac{y}{2y_1} = 1,$$

and it passes through $(6, 4)$

$$\Rightarrow \frac{6}{2x_1} + \frac{4}{2y_1} = 1$$

$$\Rightarrow \frac{3}{x_1} + \frac{2}{y_1} = 1$$

$$\text{Locus of } M \text{ is } 3x^{-1} + 2y^{-1} = 1$$

12. $(x + y + 1) + \frac{2d}{a}(y + 3) = 0$

$\therefore$ Fixed point $\equiv (2, -3)$ which lies on

$$x^2 + y^2 = 13$$

13. The equation of the line in the initial system is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

The equation of the same line after rotation of

$$\text{axes is } \frac{x}{p} + \frac{y}{q} = 1$$

Since the origin remains the same, the perpendicular distance of the line from origin must

$$\text{be unchanged. So } \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{1}{\sqrt{\frac{1}{p^2} + \frac{1}{q^2}}}$$

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$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$$

14. The point of intersection of the two lines are $(-1, -2)$.

$$\text{Distance } PM = \sqrt{10}$$

Hence the required line is one which passes through $(-1, -2)$ and is $\perp$ to P.M.

$$15. \tan \theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$

$$m = \frac{2\sqrt{\frac{25}{4} + 24}}{2} = \frac{11}{2}$$

$$2m = 11$$

$$\text{and } a^2 + b^2 - ab - a - b + 1 \leq 0$$

$$\Rightarrow (a - b)^2 + (a - 1)^2 + (b - 1)^2 \leq 0$$

$$a = b = 1$$

$$\therefore 5a + 6b = 11 = 2m$$

16. distance between parallel

$$\Rightarrow t^2 + 256 + 32t = 4t^2 - 80t + 256 \text{ where } t = a^2$$

$$t = \frac{11^2}{3} \Rightarrow a^2 = \frac{112}{3} \Rightarrow a = \sqrt{\frac{112}{3}}$$

17. $m, n \in \mathbb{N}, 2m + n - 2 > 0$ and

$$4m + 5n - 20 < 0 \text{ possible pairs are}$$

$$(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)$$

$$\text{Total} = 6$$

18. Two lines are $3x + 2y = 0$ (1)

$$\text{and } 5x + 7y = 0 \quad (2)$$

Perpendicular distances from $(2, 3)$ to (1) and (2)

$$\text{are } \frac{12}{\sqrt{13}} \text{ and } \frac{31}{\sqrt{74}}$$

$$\text{Clearly } d_1 = \frac{31}{\sqrt{74}}, d_2 = \frac{12}{\sqrt{13}}$$

$$\therefore d_1^2 - d_2^2 + \frac{1}{74} + \frac{1}{13} = \frac{961}{74} - \frac{144}{13} + \frac{1}{74} + \frac{1}{13}$$

$$= \frac{962}{74} - \frac{143}{13}$$

$$= 13 - 11 = 2$$

$$\begin{aligned}
 19. \quad & py^3 - 3xy^2 + 3x^2y + 3x^3 = 0 \\
 \Rightarrow & p\left(\frac{y}{x}\right)^3 - 3\left(\frac{y}{x}\right)^2 + 3\left(\frac{y}{x}\right) + 3 = 0 \\
 \Rightarrow & pm^3 - 3m^2 + 3m + 3 = 0 \quad (1)
 \end{aligned}$$

Where $m = \frac{y}{x}$

m_1, m_2, m_3 are roots and

$$m_1 m_2 = -1$$

But $m_1 m_2 m_3 = -\frac{3}{p}$

$$\Rightarrow -m_3 = \frac{-3}{p} \Rightarrow m_3 = \frac{3}{p}$$

m_3 is a root of (1) $\Rightarrow p = -3$

Equation of required line is $y - 1 = \frac{1}{3}(x - 1)$

i.e., $x - 3y + 2 = 0$

20. New coordinates of

$$B = \left(0 + 2\left(\frac{\sqrt{3}}{2}\right), 0 + 2\left(\frac{1}{2}\right) \right) = (\sqrt{3}, 1),$$

$$C = \left(0 + 2\left(-\frac{1}{2}\right), 0 + 2\left(\frac{\sqrt{3}}{2}\right) \right) = (-1, \sqrt{3})$$

New equation of BD is

$$y - \sqrt{3} = \frac{\sqrt{3} - 1}{-(\sqrt{3} + 1)}(x + 1)$$

$$y - \sqrt{3} = -(2 - \sqrt{3})(x + 1)$$

$$= -(2 - \sqrt{3})x - 2 + \sqrt{3}$$

i.e., $(2 - \sqrt{3})x + y = 2(\sqrt{3} - 1)$

21. $x^2 - 5x + 6 = 0$

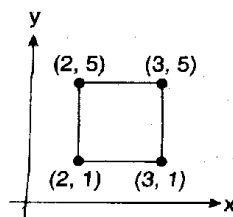
$$\Rightarrow (x - 2)(x - 3) = 0$$

$\Rightarrow$ two of the parallel sides are $x = 2, x = 3$

$$y^2 - 6y + 5 = 0$$

$$\Rightarrow (y - 1)(y - 5) = 0$$

$\Rightarrow$ other two parallel sides are $y = 1, y = 5$



So, the vertices of the parallelogram are (2, 1), (2, 5), (3, 1), (3, 5).

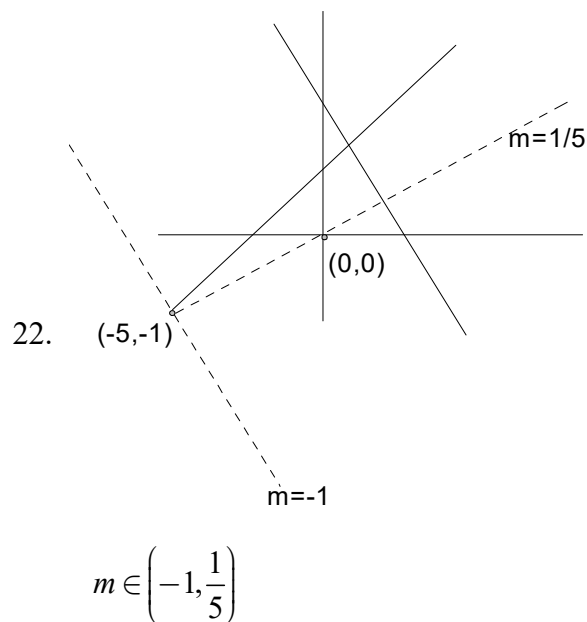
Equation of the diagonal are

$$y - 5 = \frac{1 - 5}{3 - 2}(x - 2)$$

$$\Rightarrow 4x + y = 13 \text{ and}$$

$$y - 1 = \frac{5 - 1}{3 - 2}(x - 2)$$

$$\Rightarrow y = 4x - 7$$



22. $PI = r \operatorname{cosec} 45^\circ \Rightarrow PI^2 = 2r^2$

Let I be $(\alpha, \beta) \Rightarrow (3 - \alpha)^2 + (4 - \beta)^2 = 2(|\beta|)^2$

$$\Rightarrow x^2 - y^2 - 6x - 8y + 25 = 0$$

24. $\frac{1}{2}ab = \frac{1}{2} \times h \times \sqrt{a^2 + b^2} \Rightarrow h = \frac{ab}{\sqrt{a^2 + b^2}}$

25. It is the line passing through point of intersection of diagonals and parallel to the given line

26. Area of hexagon PQRSTU

$$= \Delta - \frac{1}{9} \left(\frac{1}{2} ab \sin C + \frac{1}{2} bc \sin A + \frac{1}{2} ac \sin B \right)$$

STRAIGHT LINES

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$$= \Delta - \frac{1}{9} 3\Delta = \Delta - \frac{\Delta}{3} = \frac{2\Delta}{3}$$

$$\therefore \frac{\text{Area of hexagon } PQRSTU}{\text{Area of } \triangle ABC} = \frac{2}{3}$$

27. Let the line be $\frac{x-3}{\cos \theta} = \frac{y-4}{\sin \theta} = r$

Any point on this line be

$$(3 + r \cos \theta, 4 + r \sin \theta)$$

Substituting this in $y=8$, we get $PL = \frac{4}{\sin \theta}$

Substituting this in $x=6$ we get $PM = \frac{3}{\cos \theta}$

$$\frac{1}{PQ} = \frac{\sin \theta}{4} + \frac{\cos \theta}{3} \Rightarrow 12 = 3r \sin \theta + 4r \cos \theta$$

$$= 3(y-4) + 4(x-3)$$

$$\therefore \text{Locus is } 4x + 3y - 36 = 0$$

28. Let C be (x, y)

$$\Rightarrow \frac{x-4}{\cos 60^\circ} = \frac{y-0}{\sin 60^\circ} = 2$$

$$\therefore x = 5; y = \sqrt{3}$$

29. Note that the given family of lines are $\perp$

$$A\left(-\frac{6}{5}, \frac{-1}{5}\right); B\left(-\frac{9}{5}, \frac{-6}{5}\right)$$

Let C be any point on the circle for which AB is diameter

$$\therefore C \text{ is } \left(\frac{-15}{10} + \frac{\sqrt{34}}{10} \cos \theta, \frac{-7}{10} + \frac{\sqrt{34}}{10} \sin \theta \right)$$

$\therefore G$ is

$$\left(\frac{\frac{-45}{10} + \frac{\sqrt{34}}{10} \cos \theta}{3}, \frac{\frac{-21}{10} + \frac{\sqrt{34}}{10} \sin \theta}{3} \right) = (\alpha, \beta)$$

$$\therefore \text{Locus of } (\alpha, \beta) \text{ is } \left(x + \frac{3}{2} \right)^2 + \left(y + \frac{7}{10} \right)^2 = \frac{17}{450}$$

30. $P_1 = \left(\frac{1}{2}, 0 \right)$

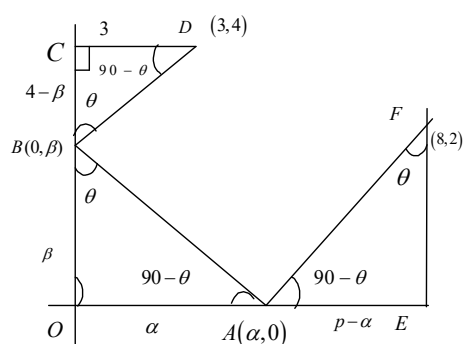
$$P_2 = \left(\frac{1}{4}, 0 \right)$$

$$P_n = \left(\frac{1}{2^n}, 0 \right)$$

$$\therefore OP_n = \frac{1}{2^n}$$

31. $\triangle BCD \sim \triangle BOA$ and $\triangle BOA \sim \triangle FEA$

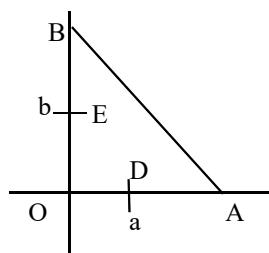
we get $\alpha = \frac{13}{3}$



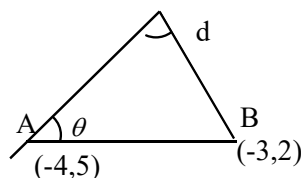
32. $AE^2 = 9 \Rightarrow a^2 + \frac{b^2}{4} = 9$

$$BD^2 = 16 \Rightarrow \frac{a^2}{4} + b^2 = 16 \Rightarrow a^2 = \frac{16}{3}, b^2 = \frac{44}{3}$$

$$\text{Area} = \frac{1}{2} ab = \frac{1}{2} \frac{4}{\sqrt{3}} \frac{2\sqrt{11}}{\sqrt{3}} = \frac{4}{3} \sqrt{11}$$



33. $d = AB \cos \theta \leq AB \therefore \text{max of } d = AB = \sqrt{10}$



34. On the line $y = 1$, the number of lattice

points is $\left[\frac{2007 - 223}{9} \right] = 198$

Total no of points

$$= \sum_{y=1}^8 \left[\frac{2007 - 223y}{9} \right] = 888$$

$$35. \quad \sqrt{2}|2x - y - 3| + |x + 3y - 4| = 7\sqrt{10}$$

$$\pm \sqrt{2}(2x - y - 3) \pm (x + 3y - 4) = 7\sqrt{10}$$

$$\text{area} = \left| \frac{(C_1 - C_2)(d_1 - d_2)}{a_1 b_2 - a_2 b_1} \right| = \frac{14^2 \cdot 10}{14\sqrt{2}} = 70\sqrt{2}$$

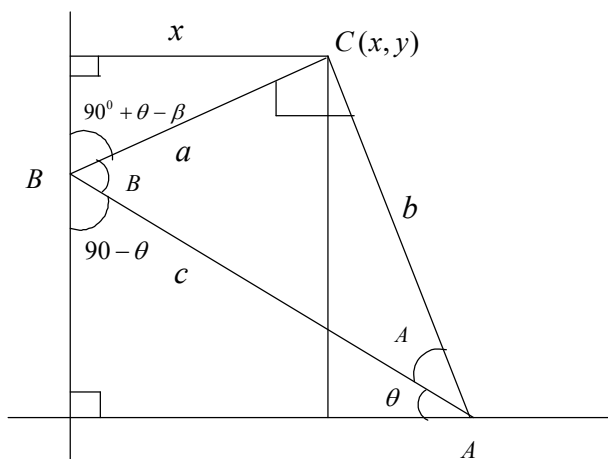
36.

$$x = a \cos(B - \theta)$$

$$= a(\cos B \cos \theta + \sin B \sin \theta)$$

$$= a \left(\frac{a}{c} \cdot \cos \theta + \frac{b}{c} \sin \theta \right)$$

$$y = b \sin(\theta + A) = b \sin \theta \cos A + b \cos \theta \sin A$$



37.

$$OA \perp BC \Rightarrow \text{slope of } OA = 1, A(\alpha, \alpha)$$

$$OB \perp CA \Rightarrow \text{slope of } OB = \frac{1}{2}, B(2\beta, \beta)$$

$$OC \perp AB \Rightarrow \text{slope of } OC = -\frac{1}{3}, C(-3\gamma, \gamma)$$

The centroid

$$G = \left(-2\gamma, \frac{-4\gamma}{9} \right) \Rightarrow \text{locus of } G \text{ is } \frac{y}{x} = \frac{2}{9}$$

38.

$$\frac{\gamma - \alpha}{\beta - \gamma} + \frac{\delta - \alpha}{\beta - \delta} = 0 \Rightarrow (a + \beta)(\gamma + \delta) = 2(a\beta + \gamma\delta)$$

$$\Rightarrow \left(\frac{-2h}{a} \right) \left(\frac{-2h_1}{a_1} \right) \Rightarrow ab_1 + a_1b = 2hh_1$$

39. Each of the lines PQ and PR is inclined at angle 45° with QR. The slope of the lines, thus

$$= \frac{-2 \pm \tan 45^\circ}{1 \mp (-2) \tan 45^\circ} = \frac{-2+1}{1+2} \text{ or}$$

$$\frac{-2-1}{1-2} = -\frac{1}{3}, 3$$

Combined equation of the lines PQ and PR is

$$[(y-1) - 3(x-2)][(y-1) + \frac{1}{3}(x-2)] = 0$$

$$\Rightarrow (y-3x+5)(3y+x-5) = 0$$

$$\Rightarrow 3y^2 - 3x^2 - 8xy + 20x + 10y - 25 = 0$$

$$\Rightarrow 3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$$

40. Mid point of Q(6, -1) and R(7, 3) is

$$\left(\frac{6+7}{2}, \frac{-1+3}{2} \right) = \left(\frac{13}{2}, 1 \right)$$

$$\text{Slope of median through P} = \frac{1-2}{\frac{13}{2}-2} = -\frac{2}{9}$$

Equation of the required line is

$$y+1 = -\frac{2}{9}(x-1) \text{ or } 2x+9y+7=0$$

41. Solving two equations $3x+4y=9$ and $y=mx+$

$$1, \text{ we get } x = \frac{5}{3+4m}$$

Now x is integer if $3+4m = 1, -1, 5$ or -5

$$\therefore m = -\frac{1}{2}, -1, \frac{1}{2}, -2$$

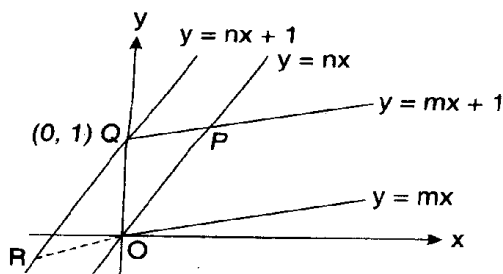
So, the integral values of m are -1 and -2 and clearly, for these values of m, x is integer

42. Area of parallelogram OPQR = 2 × area of triangle OPQ

$$\text{Coordinates of P are } \left(\frac{1}{n-m}, \frac{n}{n-m} \right).$$

$$\text{Desired area} = 2 \times \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ \frac{1}{n-m} & \frac{n}{n-m} & 1 \end{vmatrix} = \frac{1}{|n-m|}$$

STRAIGHT LINES



43. Slope of PQ = $\frac{3\sqrt{3}}{3} = \sqrt{3}$

$\therefore \tan \theta = \sqrt{3} = 60^\circ$

$\therefore \angle PQR = 120^\circ$

Bisector QS has 60° angle with RQ.

$\Rightarrow$ Slope of QS = $\tan 120^\circ = -\sqrt{3}$

and its equation is $y = -\sqrt{3}x$.

44. Let $(r_1 \cos \theta, r_1 \sin \theta)$ on $4x + 2y = 9$ then

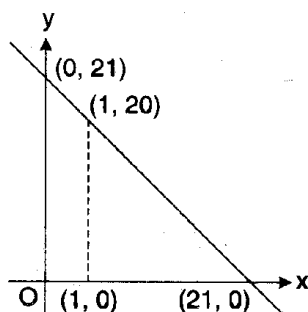
$$r_1 = \frac{9}{4 \cos \theta + 2 \sin \theta}$$

Let $(-r_2 \cos \theta, -r_2 \sin \theta)$ lies on $2x + y + 6 = 0$

Then $r_2 = \frac{6}{2 \cos \theta + \sin \theta}$

Thus the desired ratio = $\frac{OP}{OQ} = \frac{r_1}{r_2} = \frac{3}{4}$

45. Consider the line $x = 1$, which cuts the line joining points $(0, 21)$ and $(21, 0)$ at $(1, 20)$, so there are 19 integral points on this line inside the triangle. Similarly the lines $x = 2, x = 3, \dots, x = 20$ contain respectively 18, 17, ..., 0 integral points.



Total number of such points

$$= 19 + 18 + 17 + \dots + 1 = \frac{19 \times 20}{2} = 190$$

46. Area of rectangle BCDE = $4mn$.

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$$\text{Area of } \triangle ABE = \frac{2m(m-n)}{2}$$

$$= m^2 - mn$$

$$\therefore \text{area of pentagon} = 4mn + m^2 - mn$$

$$= m^2 + 3mn$$

47. Point R is the centroid of the triangle OPQ

$$\therefore R \text{ is } \left(\frac{0+3+6}{3}, \frac{0+4+0}{3} \right) \equiv \left(3, \frac{4}{3} \right)$$

($\because$ In $\triangle ABC$, with centroid G, areas of $\triangle GBC$, $\triangle GCA$ & $\triangle GAB$ are equal)

48. The slope of the given line = $-\sqrt{3}$

$\therefore$ The slope of the desired line L will be given

$$\text{by } m = \frac{-\sqrt{3} - \tan 60^\circ}{1 + (-\sqrt{3}) \tan 60^\circ} \quad \text{or}$$

$$\frac{-\sqrt{3} + \tan 60^\circ}{1 - (-\sqrt{3}) \tan 60^\circ}$$

$$= \frac{-2\sqrt{3}}{-2} \text{ or } 0$$

$$= \sqrt{3} \text{ or } 0$$

MULTI ANSWER QUESTIONS

49. Bisectors are perpendicular to each other, slope of perpendicular bisector is 2
It is passing through $(1, 2)$
Equation of other bisector is
 $y - 2 = 2(x - 1)$
i.e., $2x - y = 0$

50. $(2x + 3y - 5) \cos \theta + (3x - 5y + 2) \sin \theta = 0$
Point of intersection of $2x + 3y - 5 = 0$ and $3x - 5y + 2 = 0$ is $(1, 1)$
Let (h, k) be reflection of $(1, 1)$ in the line
 $x + y = \sqrt{2}$

$$\frac{h-1}{1} = \frac{k-1}{1} = \left(\frac{1+1-\sqrt{2}}{2} \right)$$

$$h = \sqrt{2} - 1, k = \sqrt{2} - 1$$

$$\therefore (h, k) = (\sqrt{2} - 1, \sqrt{2} - 1)$$

51. A line through $A(-5, -4)$ with slope $\tan \theta$ is

$$\frac{x+5}{\cos \theta} = \frac{y+4}{\sin \theta} = r$$

Any point on the line is

$$= (-5 + r \cos \theta, -4 + r \sin \theta)$$

If this lies on $x + 3y + 2 = 0$, we have

$$-5 + r \cos \theta + 3(-4 + r \sin \theta) + 2 = 0$$

$$\therefore r = \frac{15}{AB} = \cos \theta + 3 \sin \theta$$

$$\text{similarly, we get, } \frac{10}{AC} = 2 \cos \theta + \sin \theta$$

$$\text{and } \frac{6}{AD} = \cos \theta - \sin \theta$$

From conditions,

$$(\cos \theta + 3 \sin \theta)^2 + (2 \cos \theta + \sin \theta)^2 = (\cos \theta - \sin \theta)^2$$

$$\Rightarrow (2 \cos \theta + 3 \sin \theta)^2 = 0$$

$$\Rightarrow \tan \theta = -\frac{2}{3}$$

$$52. (3x + 4y + 5)(x + 2y + 3) = 0$$

$$m_1 = -3/4, m_2 = -\frac{1}{2} \Rightarrow m_1 m_2 = \frac{3}{8}$$

$$p \equiv (1, -2)$$

$$\tan \theta = \frac{-\frac{1}{2} + \frac{3}{4}}{1 + \frac{3}{8}} = \frac{2}{11} \Rightarrow \sin \theta = \frac{2}{5\sqrt{5}}$$

53. The diagonal of rhombus is parallel to angle bisector of given lines.

$$\therefore \text{Diagonals are } 2y + x = 5 \text{ and } 2x - y = 0$$

$$\therefore \text{Possible coordinates of } A \text{ are } \left(0, \frac{5}{2}\right) \text{ or } (0, 0)$$

54. Let BC be $x + y = 5$, then P be the point of concurrency of family of lines

$\therefore P = (1, 2)$ let A be midpoint of BC, equation of PA is $x - y + 1 = 0$; PB and PC

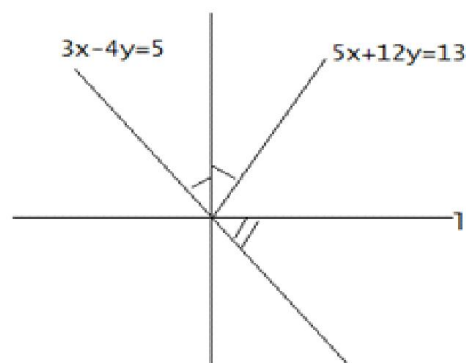
make equal angles $\frac{\pi}{6}$ with $x - y + 1 = 0$

55. Sum of roots = -8.

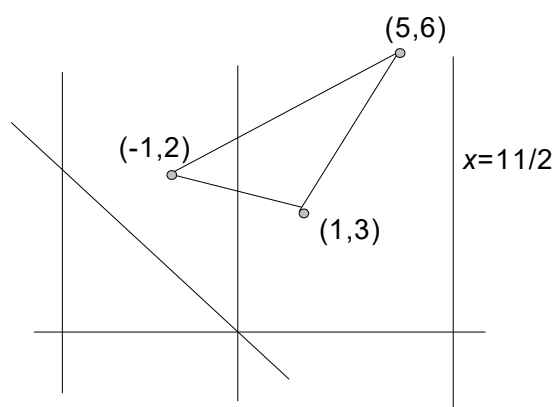
56. Triangle formed is obtuse

57. clearly, the line l can be any one of the bisectors of the angles between the lines $3x - 4y = 5$ and

$$5x + 12y = 13$$



58.



$$59. \frac{m_1}{m_2} = \frac{9}{2}, \quad \frac{m_1 - m_2}{1 + m_1 m_2} = \pm \frac{7}{9}$$

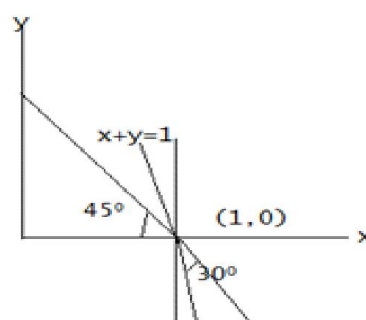
$$\Rightarrow 9m_2^2 - 9m_2 + 2 = 0,$$

$$9m_2^2 + 9m_2 + 2 = 0$$

$$m_2 = \frac{2}{3}, \frac{1}{3} \text{ \& } m_2 = \frac{-2}{3}, \frac{-1}{3}$$

60. The line of the refracted ray passes through the point (1, 0) and its slope is $\tan 105^\circ$

$$\therefore \text{The equation of the line of the refracted ray is } y - 0 = \tan 105^\circ \cdot (x - 1)$$

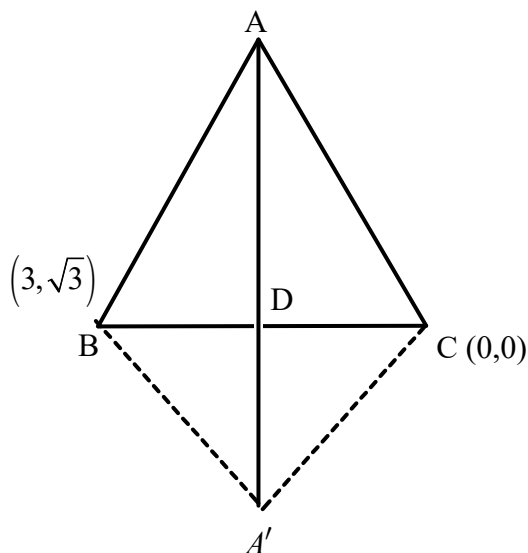


COMPREHENSION QUESTIONS

PASSAGE : I

61. $\Delta DEF = \frac{1}{4}$ area of ABC
62. Equation of perpendicular bisector of AC is
 $\Rightarrow y + 2x = 11$
 perpendicular bisector of BC is $x = 4$
63. Solving altitudes through A & B we get $(9, -5)$

PASSAGE : II (64 to 66)



C is image of B w.r.t. $\sqrt{3}x + y - 2\sqrt{3} = 0$

$A = (0, 2\sqrt{3})$, A' is image of A w.r.t. $\overline{BC}$.

Equation of altitude AD is $\sqrt{3}x + y - 2\sqrt{3} = 0$

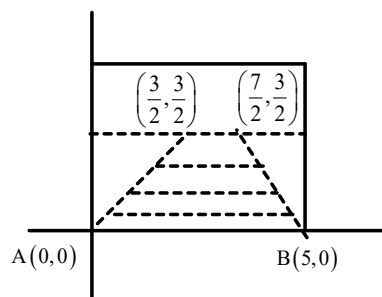
The image of point B about the line AD is $(0, 0)$

$$\therefore BC = 2\sqrt{3}$$

Clearly $(2\sqrt{3}, 0)$ is not a vertex of triangle

PASSAGE : III

67. $d(P, AB) \leq d(P, BC)$



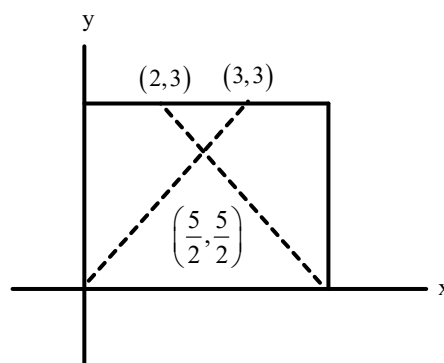
$$d(P, AB) \leq d(P, CD)$$

$$d(P, AB) \leq d(P, AD)$$

$\Rightarrow P$ lies in region as shown

$$\text{Area} = \frac{1}{2} \times \frac{3}{2} \times \frac{21}{4}$$

68. $d(P, AB) \geq d(P, BC)$



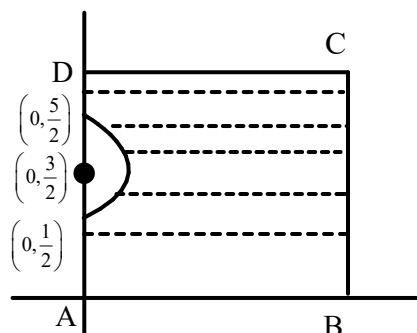
$$d(P, AB) \geq d(P, CD)$$

$$d(P, AB) \geq d(P, AD)$$

$\Rightarrow P$ lies in region as shown

$$\text{Area} = \frac{1}{2} \times 1 \times \frac{1}{2} = \frac{1}{4}$$

69. $x^2 + \left(y - \frac{3}{2}\right)^2 \geq 1$



$$\text{Area} = 5 \times 3 - \frac{\pi(1)^2}{2} = 15 - \frac{\pi}{2}$$

PASSAGE : IV (70 - 72)

$$PB = x, BQ = y, \frac{1}{2}xy \sin \theta = \frac{1}{2} \times \frac{1}{2} \times 6 \times 8$$

$$xy \frac{4}{5} = \frac{1}{2} \times \frac{1}{2} \times 6 \times 8 \Rightarrow xy = 30 \quad x + y = 12$$

$$2 \Rightarrow x + \frac{30}{x} = 12$$

$$\Rightarrow x^2 - 12x + 30 = 0$$

$$x = \frac{12 \pm \sqrt{144 - 120}}{2} = \frac{12 \pm 2\sqrt{6}}{2} = \pm\sqrt{6}$$

But

$$x \neq 6 + \sqrt{6} \Rightarrow x = 6 - \sqrt{6} \quad \therefore P = (0, \sqrt{6})$$

$$y = \frac{30}{6 - \sqrt{6}} = \frac{30(6 + \sqrt{6})}{30} = 6 + \sqrt{6}$$

$$Q.A = 10 - (6 - \sqrt{6}) = 4 - \sqrt{6}$$

$$\sin a = \frac{y}{4 - \sqrt{6}} \Rightarrow \frac{3}{5} = \frac{y}{4 - \sqrt{6}} \Rightarrow y = \frac{3(4 - \sqrt{6})}{5}$$

$$6x + \frac{24(4 - \sqrt{6})}{5} = 48 \Rightarrow x = 8 - \frac{(16 - 4\sqrt{6})}{5}$$

$$\frac{24 + 4\sqrt{6}}{5} \quad Q \left(\frac{24 + 4\sqrt{6}}{5}, \frac{12 - 3\sqrt{6}}{5} \right)$$

$$\text{Slope of line is } \frac{\frac{12 - 3\sqrt{6}}{5}}{\frac{24 + 4\sqrt{6}}{5} - 0} = \frac{12 - 8\sqrt{6}}{24 + 4\sqrt{6}} = \frac{3 - 2\sqrt{6}}{6 + \sqrt{6}}$$

$$= \frac{(3 - 2\sqrt{6})(6 - \sqrt{6})}{30}$$

$$= \frac{18 - 15\sqrt{6} + 12}{30}$$

$$= \frac{30 - 15\sqrt{6}}{30} = \frac{2 - \sqrt{6}}{2}$$

MATRIX-MATCHING QUESTIONS

73. (A) Let $A = (a, 0), B = (0, b)$

STRAIGHT LINES

$$\Rightarrow a^2 + b^2 = l^2$$

If the circumcentre is (x, y)

$$x = \frac{a}{2}, y = \frac{b}{2} \Rightarrow x^2 + y^2 = \frac{a^2 + b^2}{4} = \frac{l^2}{4}$$

B) If orthocentre is the point O, then locus contains just one point i.e., $x^2 + y^2 = 0$

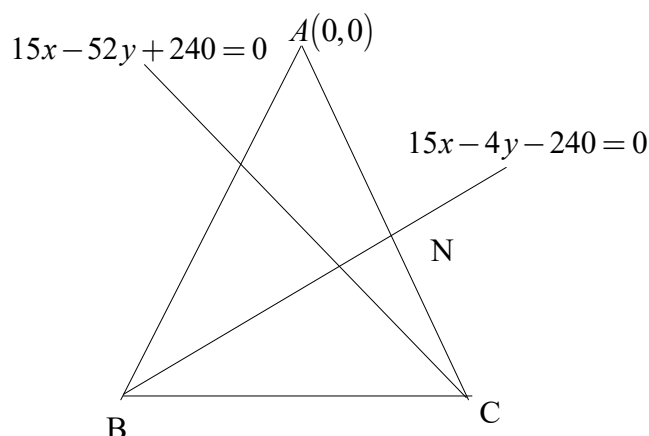
C) Let (x, y) be incentre

$$x = \frac{ab}{a + b + l}, y = \frac{ab}{a + b + l} \Rightarrow x = y$$

D) Let (x, y) be centroid $\Rightarrow x = \frac{a}{3}, y = \frac{b}{3}$

$$\Rightarrow x^2 + y^2 = \frac{l^2}{9}$$

74.



$$15x - 52y - 240 = 0 \quad 15x - 4y - 240 = 0$$

$$\text{Let } B \left(x_1, \frac{15x_1 - 240}{4} \right)$$

$$\text{and } C \left(x_2, \frac{15x_2 + 240}{52} \right)$$

Mid point of AB and AC are

$$M \left(\frac{x_1}{2}, \frac{15x_1 - 240}{8} \right) \quad \text{and}$$

$$N \left(\frac{x_2}{2}, \frac{15x_2 + 240}{104} \right) \text{ lie} \quad \text{on}$$

$$15x - 25y + 240 = 0 \quad \text{and} \quad 15x - 4y - 240 = 0$$

$$\therefore x_1 = 20 \quad \text{and} \quad x_2 = 36$$

STRAIGHT LINES

$$\therefore B(20,15) \text{ and } C(36,15)$$

$$\therefore a = BC = 16, b = CA = 39, C = AB = 25$$

$$\therefore \text{Incentre } I(21, 12), \text{ Centroid } G\left(\frac{56}{3}, 10\right)$$

excentre opposite to C, $I_3(-4, 7)$ and orthocentre is $(0, 63)$

$$75. \text{ A) } 2a^2 + a - 3 < 0 \Rightarrow (2a+3)(a-1) < 0$$

$$\Rightarrow a \in (0, 1)$$

No. of integral values of $a = 0$

$$\text{B) } a^2 + a - 2 = 0 \Rightarrow a = -2, +1$$

$$4a^2 + 4a - 3 = 0 \Rightarrow a = \frac{1}{2}, \frac{3}{2}$$

$$\therefore a \in \left(-2, -\frac{3}{2}\right) \cup \left(\frac{1}{2}, 1\right)$$

Values of a of the form $\frac{k}{3}$ are $\frac{-5}{3}, \frac{2}{3}$

C) Slope of line joining $(t-1, 2t+2)$ and

$$(2t+1, t) \text{ is } \frac{2t+2-t}{t-1-2t-1} = -1$$

$\therefore$ Slope of perpendicular bisectors is 2

D) Image of A w.r.t $y = x$ and $y = 0$ lies on BC which are $(2, 1), (1, -2)$

$$76. \text{ A) } a + c = 2b \Rightarrow a + c - 2b = 0$$

$$\Rightarrow ax + by + c = 0 \text{ Satisfies } (1, -2)$$

B) Perpendicular distance of $P(\lambda, 4-\lambda)$ from

$$4x + 3y = 10$$

$$\frac{|4\lambda + 3(4-\lambda) - 10|}{5} = 1$$

$$\Rightarrow |\lambda + 2| = 5 \Rightarrow \lambda - 2 \pm 5 = 3, -7$$

$$P(3, 1), (-7, 11)$$

C) Point $B = (-1, 2)$

$$\text{D) } \frac{y}{x} = \frac{7}{4}$$

$$\frac{4+1}{y-5} = \frac{2}{3}, \frac{21}{4}x = 2x - 13$$

$$x = -4, y = -7$$

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$$77. \text{ A) } \begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0 \Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

$$\text{B) } \begin{vmatrix} 2a & 3a & 1 \\ 3b & 2b & 1 \\ c & c & 1 \end{vmatrix} = 0 \Rightarrow bc + a - 5ab = 0$$

$$\text{C) } \begin{vmatrix} a & 2 & 1 \\ b & 3 & 1 \\ c & 4 & 1 \end{vmatrix} = 0 \Rightarrow a + c = 2b$$

$$\text{D) } \begin{vmatrix} a & a & 0 \\ c & c & b \\ 1 & 0 & 1 \end{vmatrix} = 0 \Rightarrow c^2 = ab$$

INTEGER QUESTIONS

78. Let the equation of BC be $Y=0$ then the coordinates of B are $(-36, 0)$ and of C are $(-63, 0)$, where equation of AB is $37x - 36y + 37 \times 36 = 0$ and of AC is $64x - 63y + 64 \times 63 = 0$. Equation of the altitudes through B and C are

$$y = \frac{-63}{64}(x + 36), y = \frac{-36}{37}(x + 63)$$

Solving those equations we get the coordinates of the orthocentre of the triangle ABC

$$\text{So, } a = 36 \times 63, b = -36 \times 63$$

$$\Rightarrow a - b = 2 \times 36 \times 63 = 4536$$

$$79. \text{ Minimum area of } \triangle OPQ = P = 2 \times 87 \times 33 \\ P - 5740 = 2.$$

Minimum area of Triangle formed by a line passing through (a, b) with co-ordinate axis $= 2ab$

$$80. \text{ } a = 12, h = -5, g = \frac{11}{2}, b = 2, f = -\frac{5}{2}, c = \lambda$$

$$\begin{vmatrix} 2 & -5 & \frac{11}{2} \\ -5 & 2 & -\frac{5}{2} \\ 11/2 & -5/2 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 2$$

81. A and B are on opposite sides of y - axis. Reflecton of A $(-3, 2)$ is y - axis is $A' = (3, 2)$

so tht A', B will be on the same side of P

Equation of A'B is $y - 2 = -3(x - 3)$

$$\Rightarrow y - 2 = -3x + 9$$

$$3x + y = 11$$

This meets y-axis at Q (0, 11)

Q is required position of P so that $k = 11$

$$\therefore \frac{k-1}{5} = \frac{10}{5} = 2$$

(Since difference of two sides of a triangle is smaller than the third side)

Note: We know that $|PA - PB|$ will be greatest if P lies on the line containing A and B, rather A and B must lie on the same side of P.

82. Let $L : y = mx$

$$\therefore P\left(\frac{1}{m+1}, \frac{m}{m+1}\right) \text{ and } Q\left(\frac{3}{m+1}, \frac{3m}{m+1}\right)$$

Since L_1 and L_2 are parallel to $y - 2x + 5 = 0$ and $3x + y - 5 = 0$ respectively

$$L_1 : y - 2x = \frac{m-1}{m+1}$$

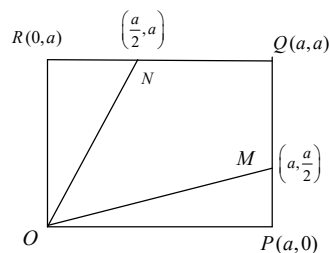
$$L_2 : y + 3x = \frac{3m+9}{m+1}$$

By eliminating m from L_1 and L_2 we get

$$\text{Locus R as } x - 3y + 5 = 0$$

$$\therefore \lambda = 5$$

83. **The centroid of the triangle and the fourth vertex of the three parallelograms formed by the given points will be required number of points (taking the order of vertices into consideration)**



84.

$$\Delta OMN = \frac{1}{2} \left| \frac{a^2}{4} - a^2 \right| = \frac{1}{2} \frac{3a^2}{4}$$

$$\frac{a^2}{\Delta OMN} = \frac{8}{3} = \frac{P}{Q}$$

85. **The line segments joining the mid points of sides are equidistant from the vertices**

86. **Locus of P is a circle with radius 2 units**

$$\therefore \text{Maximum area} = \frac{1}{2} \times 3 \times 2$$

87. Vertices of a square are

$$\left(3 + 2\sqrt{2} \left(-\frac{1}{\sqrt{2}} \right), 7 \pm 2\sqrt{2} \left(\frac{1}{\sqrt{2}} \right) \right)$$

88. Let the variable straight line be $ax + by + c = 0$, then according to question

$$\frac{2a+c}{\sqrt{a^2+b^2}} + \frac{2b+c}{\sqrt{a^2+b^2}} + \frac{a+b+c}{\sqrt{a^2+b^2}} = 0 \Rightarrow a + b + c = 0$$

So, $ax + by + c = 0$ passes through (1, 1).

89. The equations of the rays are $x + y = |a|$ and $ax - y = 1$.

$$\text{Solving we get } x = \frac{1+|a|}{1+a} \text{ and } y = \frac{a|a|-1}{1+a}$$

For the first quadrant $x > 0$ and $y > 0$

$$\Rightarrow \frac{1+|a|}{1+a} > 0 \text{ and } \frac{a|a|-1}{1+a} > 0$$

$$\Rightarrow 1 + a > 0 \text{ and } a|a| - 1 > 0$$

$$\text{If } -1 < a < 0 \text{ then } -a^2 - 1 > 0$$

$$\Rightarrow a \in \phi$$

$$\text{If } a \geq 0 \text{ then } a^2 - 1 > 0$$

$$a \geq 0 \text{ and } a > 1$$

Thus $a \in (1, \infty)$.

90. Since B (2, -7) does not satisfy the equations $3x + y + 11 = 0$ and $x + 2y + 7 = 0$. The altitude and median are drawn from A and C (or from C and A) respectively. If $C = (h, k)$ then C lies on $3x + y + 11 = 0 \Rightarrow 3h + k + 11 = 0$
Mid point D of CB lies on $x + 2y + 7 = 0$

STRAIGHT LINES

$$\Rightarrow \frac{h+2}{2} + 2\left(\frac{k-7}{2}\right) + 7 = 0,$$

Then $C = (-4, 1)$

Equation of AB is

$$y + 7 = \frac{1}{3}(x - 2) \Rightarrow 3y - x = -23$$

Now, as AC passes through the intersection of AB and median AD, its equation is of the form $3y - x + 23 + \lambda(x + 2y + 7) = 0$ and it passes through $C = (-4, 1)$, we get $\lambda = -6$

Equation of AC is $7x + 9y + 19 = 0$
 $\therefore \lambda = 19$

91.

$$PA^2 + PB^2 + PC^2 = 3PG^2 + GA^2 + GB^2 + GC^2 = 6R^2$$

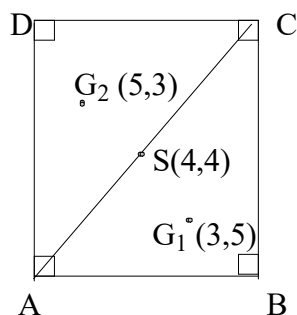
$$= \left(\therefore R = \frac{a}{2 \sin A} = \frac{\sqrt{3}}{3} = 1 \right)$$

92. It is the altitude to the least side

93. $x^2 + y^2$ is the square of perpendicular distance from origin

94. (n, n^2) lies on parabola $x^2 = y$ which intersect the line $3x + y - 4 = 0$ at $(1, 1)$ and $4x + y - 21 = 0$ at $(3, 9)$
 $\therefore$ Interior point is $(2, 4)$ which lies on parabola.

95.



B, D are ortho centres of $\triangle ABC, \triangle ADC$

Respectively, $S(4, 4)$ is circum centre of

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$\triangle ABC, \triangle ADC$,

$$DS = 3\sqrt{2} \quad ; \quad BD = 6\sqrt{2}$$

$$AB = \frac{BD}{\sqrt{2}} = 6$$

96. The minimum value of

$\sec^2 \theta + 2 \operatorname{cosec}^2$ is $(1 + \sqrt{2}) = 5.8$ but

perpendicular distance $= \frac{7}{5} < 5.8 \therefore$ No of points = 2

ASSERTION & REASON QUESTIONS

$$97. S_1 : x = \frac{3 \times 9 - 1 \times 5}{3 - 1} = \frac{22}{2} = 11$$

$$y = \frac{3 \times 6 - 5(-2)}{3 - 1} = 14$$

point is $(8, 4)$ is wrong

$\therefore S_1$ is false

$$S_2 : ax + hy + g = 0$$

$$hx + by + f = 0$$

$$gx + fy + c = 0$$

$$\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$$

$\therefore S_2$ is true

$$98. L_{11} \text{ of origin} = 2$$

$$L_{11} \text{ of } (-4, 1) = -6$$

$(-4, 1)$ & origin lie on opposite side

$$99. \text{ since } \frac{2+3+7}{3} = 4 \text{ and } \frac{3+4+7}{3} = 5$$

$\therefore$ Centroid of triangle ABC is $D(4, 5)$

$\therefore$ A, B, C, D are non-collinear points.

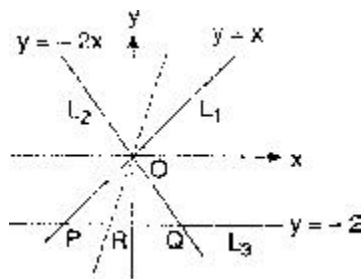
$$100. \text{ Coordinates of P are } (-2, -2)$$

Coordinates of Q are $(1, -2)$

$$\therefore OP = 2\sqrt{2} \text{ and } OQ = \sqrt{5}$$

$$\text{Now, } \frac{PR}{RQ} = \frac{OP}{OQ} = \frac{2\sqrt{2}}{\sqrt{5}}$$

$\Rightarrow$ Statement-1 is correct, however the triangles OPR and ORQ may not be similar.



$\Rightarrow$ Statement-2 is false.